

Pitch Script — Slides 1–6

SaResQ · SIH26177 · verbatim script, delivery cues, and slide-vs-reality notes · target ~4 min 30 s total

HOW TO USE THIS

Each slide has: a target duration, the script broken into beats (say it in your own cadence, not word-for-word if that feels stiff), a

CUE

line telling you what to point at or do with your hands, and — where the PPT's wording is slightly ahead of what's actually built — a note on how to say it so it stays **true to the current architecture** without contradicting the slide on screen. Practice this out loud twice before tomorrow; the pauses matter more than the words.

Slide 1 — Title

~15 s

SAY

"Good [morning/afternoon]. We're **Team SaResQ**, presenting on Problem Statement 26177 — an AI-powered drone payload for search and rescue. I'm [name], and this is [team members, if presenting jointly]."

CUE — stand centred, let the title slide sit for a beat before you speak. Don't rush the first line; a confident pause here sets the tone for the whole pitch.

SAY

"In the next four minutes, we'll show you a system that's not a concept — it's built, it's running on real hardware right now, and every number we give you today was measured on that hardware."

CUE — this line is your thesis for the whole talk. Say it slowly.

Slide 2 — Problem & Solution

~50 s

SAY — THE PROBLEM

"After a disaster, the first few hours decide outcomes. But three things slow rescue teams down. **Limited visibility** — survivors are hard to spot from the air, especially at night. **Difficult terrain** — the ground is often too damaged or dangerous to search on foot. And **manual analysis** — someone still has to sit and watch hours of drone footage, and a tired human misses things."

CUE — point to the three PROBLEM bullets as you say each one.

SAY — THE SOLUTION

"SaResQ automates the watching, not the flying. It's a payload that bolts onto any drone a rescue team already owns, and does three things onboard, in real time, with no internet connection: it runs **thermal-first detection** — a lightweight cascade finds body heat first, then confirms with a vision model, so we never need a heavy network running on every single frame. It **classifies the disaster scene** — fire, flooding, collapsed structures — so responders get context, not just a person-count. And it **fuses multiple observations** — heat, motion, and vision — into one confidence score per detection, so a coordinator knows not just 'something's there' but how sure we are, and why."

CUE — point to the three SOLUTION bullets in the same order as the problem bullets, so the audience sees problem→answer pairs.

SLIDE-VS-REALITY NOTE

The slide says "detects survivors using lightweight YOLOv8n" — true, but say it as part of the cascade, not standalone: the network only ever looks at the small patches the thermal stage flags. That's the actual innovation, and it's what makes this run on a ₹4,000 computer instead of needing a GPU.

SAY — HARDWARE

"On the hardware side: a Raspberry Pi 4 as the onboard computer, an MLX90640 thermal array, a Pi Camera for the visible spectrum, and GPS plus an IMU for position and orientation. All commodity parts, all together under ₹14,000."

CUE — point to the Hardware & Sensors column.

SAY — THE AI PIPELINE, IN ORDER

"Here's how a frame actually moves through the system. The thermal sensor reads 32-by-24 pixels, four times a second. A statistical gate checks every single one of those frames for free — it's just arithmetic, no neural network yet — and flags the handful of warm or cold patches worth a closer look. **Only those small crops** go to YOLOv8n, our detector, which confirms whether it's actually a person. In parallel, MobileNetV2 looks at the full frame and classifies the disaster scene — fire, flood, collapsed building — at over 95% accuracy. Then our fusion layer combines heat, motion, and the visual confirmation into one ranked verdict per detection, so responders see a confidence level and the reason behind it, not just a box on a screen."

CUE — trace the pipeline left to right with your hand: gate → crop → detector → fusion. This is the moment to slow down; it's the technical heart of the pitch.

SAY — WHY IT'S EFFICIENT

"Everything is quantised to run without a GPU — INT8 weights on a TFLite runtime — and the whole pipeline, worst case, takes 177 milliseconds on the Pi, against a 250 millisecond budget. That headroom is measured, not estimated."

CUE — point to INT8/TFLite on the slide.

SAY — SOFTWARE

"On the ground side: Flask serves a live dashboard, SQLite stores every detection and its evidence, and Leaflet renders the live map — so a coordinator sees detections appear near the drone's position as they're confirmed, in real time, fully offline."

CUE — point to the Software column, then gesture toward the demo device if you have it live.

SLIDE-VS-REALITY NOTE

"RGB–Thermal Fusion" on the slide is our verdict ladder — a transparent, ranked combination of heat + motion + vision, not a black-box network. Say it as "our fusion layer" confidently; if a judge asks why not a learned fusion network, the honest answer is in the Q&A sheet — calibration and explainability both mattered more than squeezing out another point of accuracy.

SAY — OPERATIONAL

"This is airframe-agnostic — it deploys on any F450-class commodity drone a department already flies, with zero electrical connection to the flight controller. It's fully offline by design: detection, classification, and geo-tagging all happen onboard, no connectivity required. And we validated it in the right order — characterised on a fixed bench first, then flight-tested — specifically so vibration and wind don't get confused with the AI's own accuracy."

CUE — point to Operational Feasibility column.

SAY — FINANCIAL

"The AI and search-and-rescue system — the part we built — comes to about ₹13,000. A complete autonomous drone around it is roughly ₹30,000. That's seven to ten times cheaper than enterprise thermal SAR drones on the market today, which run three to twenty-two lakh rupees. And training cost us nothing — we used free compute."

CUE — point to the ₹13,000 / ₹30,000 numbers; let them land, don't rush past them.

SAY — TECHNICAL FEASIBILITY

"This isn't a guess at what might work. It's built on a peer-reviewed reference result — a 2018 study that got over 99% CNN accuracy on exactly this sensor combination, low-cost RGB and thermal on a Raspberry Pi. What that same study found is that running a CNN on embedded hardware in real time was, in their words, impractical. Our cascade design is the answer to that specific limitation — we don't run the network on every frame, only where the thermal gate says something's there. And because our gate checks for both hot **and** cold anomalies, we don't fail on hot rooftops and debris in direct sun the way a simple 'find the hot spot' approach would."

CUE — this is your strongest credibility line. Say "peer-reviewed" and "impractical" clearly — it shows you know the literature, not just your own build.

SAY — RISK & VALIDATION

"Flight control itself uses a proven, open-source autopilot stack — we didn't reinvent that. Our job was the perception payload, and we validated it end to end on the actual target hardware, with a working prototype you can hold."

CUE — if you have the physical prototype, this is the moment to hold it up.

Slide 5 — Impact & Benefits

~50 s

SAY — WHO BENEFITS

"Four groups benefit directly. Rescue teams get faster survivor identification and location. Disaster response agencies get better aerial awareness over large search areas. Survivors themselves have a higher chance of being found at night or in low visibility, which is exactly when human search is weakest. And drone operators do less manual video monitoring, because the AI is doing the watching."

CUE — point to each of the four WHO BENEFITS entries as you name them, quick pace here.

SAY — WHERE THE VALUE SHOWS UP, AND WHY IT'S DIFFERENT

"The value shows up as faster search — the AI prioritises where survivors are likely to be, instead of scanning everything equally. Day and night detection, because thermal doesn't need light. Accurate location sharing, because every detection is geo-tagged, not just described. And low-cost deployment, because the whole onboard brain is a Raspberry Pi."

"What makes this different isn't one clever trick — it's that we built for Indian field conditions specifically: our thermal gate is tuned to keep working in high ambient temperatures, where a naive 'hot-spot' detector would fail. It's low latency because everything runs at the edge. And it's cost-effective because the core AI system was built for about ₹13,000."

CUE — slow down on "built for Indian field conditions" — that's your localisation answer if anyone asks "why not just use an existing foreign system."

Slide 6 — Research & References

~35 s

SAY

"We didn't build this in isolation. Four papers shaped our design: ETH Zurich's work on RGB-plus-infrared detection and tracking for UAVs, flight-tested with Switzerland's real air-rescue service — that's where our re-identification and tracking ideas come from. A 2018 study on low-cost RGB-thermal pedestrian detection on embedded hardware, which we just discussed — that's our baseline. A transformer-based fusion network for multispectral victim detection at different scales. And COXNet, on cross-modal fusion for tiny-object detection from drones."

CUE — point to each numbered reference as you name it, don't linger.

SAY — THE HONEST CLOSE

"The gap all four of these papers point at is the same one we set out to solve: RGB-plus-thermal fusion works well in research, but small targets and real-time detection on low-cost hardware remain hard. That gap is exactly where SaResQ sits — and everything you saw in this presentation was measured on the actual hardware, not simulated. Thank you — happy to take questions."

CUE — end standing still, hands open, eye contact on the judges — this is the line that invites questions instead of bracing for them.

If you only remember one delivery note

Slow down on numbers and speed up on background. A judge remembers the number you paused on, not the one you rushed through.